AIGym

System Design Description

Abstract
This document describes the design and implementation of the AIGym support system of Arrowhead (documentation edition 5.3.0) (`ah5-AIGym-java-spring`).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the AIGym system implementation, which realizes the AIGym SysD [1]. AIGym is an experimentation system for machine learning in the local cloud: it collects training samples, predicts with k nearest neighbours and consumes the `sysMonitorData` stream. In Section 3, we describe implementation details of the system.

3 Implementation

This implementation is based on the SysD document AIGym [1] and is located in `ah5-AIGym-java-spring`.

3.1 Implementation language and tools

- Java 21, Spring Boot (web), Eclipse Paho MQTT, RabbitMQ AMQP client.
- Storage: in memory.

3.2 Independent source and build ownership

The project folder owns its application source, Maven build, tests, executable JAR, Dockerfile, configuration and Compose definition. No source or image is taken from a sibling support folder or JavaSupportRuntime. Build with Java 21 and `mvn -B clean verify`; a populated external Maven dependency cache permits offline builds.

AIGym retains its independent Spring Boot parent and application source. It does not use the Arrowhead common runtime or another support-system artifact.

The packaged artifact is `arrowhead-ai-gym-5.2.2.jar`. The image builds and tests this folder only; the system has a fixed application main class.

3.3 Functional properties implementation

The following components implement the system:

Component	Responsibility
`AiGymController`	HTTP operations.
`ModelStore`	k-NN store.
`AiGymRuntime`	MQTT intake and sysMonitor.
`ArrowheadGateway`	Registration, identity, orchestration, authorization.
`ArrowheadSecurityFilter`, `CrlRevocationConfig`	Caller authentication and CRL.

Every request passes `ArrowheadSecurityFilter` (identity token verified with Authentication and matched to the certificate; ConsumerAuthorization `verify` for the operation) before the controller.

3.3.1 Data handled and stored

Models and observations in memory.

The data models of the interfaces are Java records (of `eu.arrowhead.dto` in the common library, or of the module); the IDD documents name them without the `DTO` suffix. The records used by this system are: `Prediction`, `PredictionRequest`, `Sample`, `SampleRequest`.

3.4 Non functional properties implementation

3.4.1 Security

Spring TLS with mandatory client certificates and `CrlRevocationConfig`; `ArrowheadSecurityFilter` binds the identity token to the certificate.

3.4.2 Power management

None; the system is a server process without power states. The `sysMonitor` `power` observation always reports line supply.

3.4.3 Internal monitoring

Logs go to stdout. The system publishes its `sysMonitor` observations every 30 s (see the sysMonitor IDD).

3.4.4 Configuration

Configuration is read from environment variables (listed in the SysD); secrets are read from files (`*_FILE` variables or mounted credential bundles).

3.5 Deployment

The local `compose.yaml` selects only this support system. It joins an operator-selected external cloud network; it does not start core systems or another support system. Copy `.env.example`, provision this system's own identity and credential files, validate the Compose definition, then run `docker compose up -d -build` from this folder. The image uses the secure external `config/application.properties`, mutual TLS, outsourced authentication, certificate/token binding and a mounted CRL. Secrets are read through Spring's file configuration tree. The app runs as UID 10001 with a read-only root filesystem and bounded non-executable temporary/log storage.

Registry, Authentication, DSO, ConsumerAuthorization and required brokers/providers remain network dependencies. Independent deployment does not remove those service contracts. The listener health check is only a liveness check; inspect successful application startup, registration and authenticated service access before acceptance. The local `DEPLOYMENT.md` supplies the complete procedure. The previously recorded live-cloud deployment is historical baseline evidence; the new packaging does not change a live cloud.

3.6 Local modifications

Notes on this implementation in the workspace:

- Workspace system (5.2.2 naming).

3.7 Verification

Independent folder-only Maven builds run the project's JUnit suite. Artifact inspection checks the system main/source and absence of JavaSupportRuntime or other support-system implementations. Images and project-local Compose definitions are tested with private disposable state and generated test identities. Spring startup and authenticated HTTPS requests use isolated mutual-TLS core-service fixtures; missing-token and missing-certificate requests are refused. The distribution verification report records exact results and limitations. Broker, NTP and inter-cloud acceptance remain separate from deployment/startup validation.

4 Services

The implementation of the services is based on the SD and IDD documents referenced below.

Services produced	**SysD ref**	**SD ref**	**IDD ref**
`aiGym`	[1]	[2]	[3]
`sysMonitor`	[1]	[4]	[5]
Services consumed	**SysD ref**	**SD ref**	**IDD ref**
systemDiscovery, serviceDiscovery	[6]	[7]	–
identity	[8]	[9]	–
authorization	[10]	[11]	–
serviceOrchestration	[12]	[13]	–
sysMonitorData	[14]	[15]	–
sysMonitor (publisher)	AMQP broker (SysMonitor)	–	–

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-AIGym-go`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. Ordinary native roles use independent identities and JSON snapshot state rather than the Java database. Do not assume Java SQL migration, clustering or identical transaction/retry semantics. The registry has its own configuration and optional native AMQP monitor; other roles must be checked against their own registrations.

Listening port. Built-in default 127.0.0.1:8478 (Arrowhead default). The shipped config.json, compose.yaml and Dockerfile use the same port on all interfaces ("listen": "0.0.0.0:8478"; 8540–8555 layout before 2026-11-09). Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host.

Native behavior and outstanding work. Persistent samples, Java-style nearest-neighbour prediction, status. Complete observation-publishing behavior; local extension rather than upstream mandatory system. These are the boundaries recorded by the native compatibility document; later implementation changes still require direct qualification. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-AIGym-go/documentation/AIGym`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-AIGym-rust`.

The per-system entry point is declared as a binary of `arrowhead-support-rust` in `ah5-support-runtime-rust/Cargo.toml`. Build from the primary Rust workspace with the applicable package/binary

selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

The ordinary native profile uses Tokio/Axum and rustls with peer-identity checks and independent locked JSON state. Network effects are not distributed database transactions. The precise trust, CRL reload and persistence guarantees are those of the native profile, not the Java implementation.

Listening port. Default 127.0.0.1:8478 (Arrowhead default, role aigym). Override with "listen": "host:port" in the configuration, SERVER_PORT=N or –server.port N; the port overrides keep the configured host.

Native behavior and outstanding work. Persisted samples, kNN regression, observation history. Broker publishing; local extension only. These are the boundaries recorded by the native compatibility document; later implementation changes still require direct qualification. Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/AIGym`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] J. Delsing, “AIGym – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-support-java-spring/AI-gym/documentation/v5.3.0/AIGym-v5.3.0-SysD.pdf.

[2] ——, “aiGym – Service Description (AIGym),” Eclipse Arrowhead, SD 5.3.0, 2026, aIGym-v5.3.0-aiGym-SD.pdf.

[3] ——, “aiGym HTTP/TLS/JSON – Interface Design Description (AIGym),” Eclipse Arrowhead, IDD 5.3.0, 2026, aIGym-v5.3.0-aiGym-HTTP-IDD.pdf.

[4] ——, “sysMonitor – Service Description (AIGym),” Eclipse Arrowhead, SD 5.3.0, 2026, aIGym-v5.3.0-sysMonitor-SD.pdf.

[5] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (AIGym),” Eclipse Arrowhead, IDD 5.3.0, 2026, aIGym-v5.3.0-sysMonitor-AMQP-IDD.pdf.

[6] ——, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.3.0/ServiceRegistry-v5.3.0-SysD.pdf.

[7] ——, “systemDiscovery – Service Description (ServiceRegistry),” Eclipse Arrowhead, SD 5.3.0, 2026, ah5-core-java-spring/serviceregistry/documentation/v5.3.0/ServiceRegistry-v5.3.0-systemDiscovery-SD.pdf.

[8] ——, “Authentication – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-core-java-spring/authentication/documentation/v5.3.0/Authentication-v5.3.0-SysD.pdf.

[9] ——, “identity – Service Description (Authentication),” Eclipse Arrowhead, SD 5.3.0, 2026, ah5-core-java-spring/authentication/documentation/v5.3.0/Authentication-v5.3.0-identity-SD.pdf.

[10] ——, “ConsumerAuthorization – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-core-java-spring/consumer-authorization/documentation/v5.3.0/ConsumerAuthorization-v5.3.0-SysD.pdf.

[11] ——, “authorization – Service Description (ConsumerAuthorization),” Eclipse Arrowhead, SD 5.3.0, 2026, ah5-core-java-spring/consumer-authorization/documentation/v5.3.0/ConsumerAuthorization-v5.3.0-authorization-SD.pdf.

[12] ——, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-core-java-spring/serviceorchestration-dynamic/documentation/v5.3.0/DynamicServiceOrchestration-v5.3.0-SysD.pdf.

[13] ——, “serviceOrchestration – Service Description (DynamicServiceOrchestration),” Eclipse Arrowhead, SD 5.3.0, 2026, ah5-core-java-spring/serviceorchestration-dynamic/documentation/v5.3.0/DynamicServiceOrchestration-v5.3.0-serviceOrchestration-SD.pdf.

[14] ——, “SysMonitor – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-SysMonitor-java-spring/documentation/v5.3.0/SysMonitor-v5.3.0-SysD.pdf.

[15] ——, “sysMonitorData – Service Description (SysMonitor),” Eclipse Arrowhead, SD 5.3.0, 2026, sysMonitor-v5.3.0-sysMonitorData-SD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-25	5.2.2	Self-contained build, private library source, configuration, images, database and isolated deployment verification.	Codex (draft)
3	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
4	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	